Agentic FEA of the Zhangjinggao Construction Catwalk:
Coordinate gate, 142-portal reconciliation, and a CalculiX 2.21 initial-stress parse failure on a frozen hashed deck

Run `catwalk-main-deck-gate-f23d`
Repository `Hiram-test/model`, branch `cursor/catwalk-main-deck-gate-f23d`
Pull request #19

2026-08-25

Abstract

This paper records a complete, auditable agentic finite-element modelling run for the Zhangjinggao construction catwalk. A 77 MB millimetre centre-line STEP is parsed, mapped into the project convention $x = \text{chainage} - K16{+}876.000$, classified against drawing topology, coarsened, and written as a CalculiX deck. Floor-rope and portal-rope anchors are separate `*NSET`/`*BOUNDARY` families. Twenty-one cross-passages and 142 portal frames (71 stations per deck; Chinese classifier U+6980, not the look-alike U+69EC) reconcile against the drawing station lists with zero insertions. The frozen main deck `catwalk-fem/artifacts/zjg_catwalk_coarsened.inp` has SHA-256 `82548e6a3bd2612b6b39a08c313402b32a1961af6eba018158267906276ab6da` and is not rewritten. CalculiX 2.21 was executed on a byte-identical copy. Reading `*INITIAL CONDITIONS, TYPE=STRESS` failed on the first data row `E_FLOOR_ROPE,3.549611E+08` with exit code 201. No `.frd` was written; the `.dat` is empty; `.sta` and `.cvg` contain headers only; wall-clock time was 0.74–0.82 s; no assembled equation count was issued. The card is ELSET name plus one uniaxial stress. CalculiX 2.21 §7.76 requires element number, integration point, and six global stress components. Twenty-six coordinate/topology gates PASS. Frequency targets remain isolated. This paper therefore certifies a hashed, coordinate-gated, topology-reconciled deck and a documented solver-parse failure—not a solved spectrum and not a fourteen-mode match.

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1 Introduction

Related work. Automated CAD-to-FEM pipelines exist for buildings and bridges, but they usually inherit an already-consistent coordinate system and a single support family. Construction catwalks of a multi-span suspension bridge break that assumption: two walkways, two rope families with different physical anchor stations, discrete portal frames, and discrete cross-passages, all written in highway chainage. A companion theory archive (`catwalk-theory/thirteen-mode-models`, v1.2) already freezes the drawing topology and the isolation protocol for fourteen low-order families. That archive is a theoretical model family, not a CalculiX deck.

Method. The run follows `catwalk-fem/SKILL.md` and `PLAN.md`. Geometry is STEP-only. Materials, sag, and loads come from drawings and the check report. The coordinate gate is a hard stop: identity unless saddle-height evidence requires a shift, and never $X - x_{\min}$. Topology is reconciled against frozen drawing lists: 21 passages and $71 \times 2 = 142$ portal frames. Anchors are two families, not one mixed set. The writer emits a complete keyword deck. After the hash is frozen, CalculiX may be run only on a copy.

Experiment. Release `catwalk-attachment23-v2.0-s10-20260716` file `cw_S10_0716t050342_a4_centerlin` (SHA-256 `d03d01e38b823df5af4c1ff9b0b175fdfb87b097b9cda9a03af5d14e9c763344`, 139 991 trimmed curves, `SI_UNIT(.MILLI.,.METRE.)`) is converted to metres. After identity mapping, $X \in [0, 4270.609]$ m and $Z_{\max} = 350.312$ m. The written deck has 51 896 nodes and 30 317 elements (7 702 117 bytes). CalculiX CrunchiX 2.21, executed on a copy of hash `82548e6a`, dies while reading the first initial-stress row.

Outlook. A later run may emit a CalculiX-legal stress card without silently mutating this hash. North physical anchors lie outside this STEP and remain proxies. Fourteen-mode comparison stays forbidden until a solved spectrum exists.

2 Related work and isolation protocol

Related work. Agentic FEA here means a scripted, auditable chain with frozen constants, not a chat that invents sections. Dhondt's CalculiX is used as the open solver because its keyword deck is inspectable and its version is pinned. Abaqus-like ELSET-wide uniaxial prestress is a common authoring convenience; it is not the CalculiX 2.21 contract.

Method. Isolation rules, taken from theory v1.2 §1 and the skill file:

1. STEP supplies coordinates only.
2. Properties come from DRW-A/B, CALC-INPUT, STD, or a named ASSUMP.
3. `TARGET-FREQ` lives in `catwalk-fem/isolated/` and is not imported by `write_inp` or the solver deck.
4. Formed sag is $h = 227.300\,\mathrm{m}$ $(340.600 - 113.300)$. The 255.56 m main-cable control sag is forbidden in the deck (verified absent).
5. The two decks keep independent degrees of freedom; no mirror constraint.
6. Floor-rope anchors and portal-rope anchors are never unioned.
7. After SHA-256 `82548e6a` is frozen, the `.inp` is not rewritten to "make the solver happy".

Experiment. The isolated file lists fourteen frequencies starting at 0.0296 Hz. A full-text search of the frozen deck finds neither 0.0296 nor 255.56. `write_inp.py` does not import `isolated/TARGET-FREQ.json`. The STEP sibling `cw_S10_0716t050342_a4_eq.db` is listed as a forbidden source in the heading.

Outlook. Isolation is a scientific claim, not a courtesy. Any later comparison table must load TARGET-FREQ only after the deck, the solve, and the mode-shape classification are frozen.

3 Charter, evidence grades, and frozen constants

Related work. Evidence grades follow theory v1.2: DRW-A/B, CALC-INPUT, STD, TEST, ASSUMP. Portal and passage stations are DRW-B dimension chains. Saddle coordinates are CALC-INPUT (check-report tables 1-5 / 1-9).

Method. The charter (`artifacts/analysis_charter.json`) freezes: project `zjg-catwalk`, run `catwalk-main-deck-gate-f23d`, SI internal units, and the four load cases LC-DEAD-PRESTRESS, LC-PERSONNEL-UNIFORM, LC-WIND-Y, LC-FREQ. The coordinate origin is the nominal four-span north end, not a physical anchor:

$$x = \text{chainage} - K16 + 876.000, \qquad y \text{ upstream} \to \text{downstream}, \qquad z \text{ up}.$$

Physical anchors stay in four named stations:

$$\begin{aligned} x_{\text{floor,N}} &= -23.895 \text{ m } (K16 + 852.105), & x_{\text{floor,S}} &= 4210.368 \text{ m } (K21 + 086.368), \\ x_{\text{portal,N}} &= -44.909 \text{ m } (K16 + 831.091), & x_{\text{portal,S}} &= 4225.700 \text{ m } (K21 + 101.700). \end{aligned}$$

Nominal four-span breaks $\{0, 660, 2960, 3677, 4180\}$ and physical saddles $\{666.679, 2953.321\}$ are registered together and are never silently mixed.

Experiment. The heading of the frozen deck states the convention in plain text. Node $x_{\min} = 0$ equals the transformed geometry $x_{\min}$. South floor and south portal selected means differ by 11.1 m.

Outlook. If a later STEP covers the north physical anchors, those four stations must still remain four sets.

4 Ingestion and classification (N02–N07)

Related work. A centre-line STEP of 139 991 trimmed curves is not a member-role model. Role assignment has to be recovered from geometry against drawing lattices.

Method. `parse_step` stream-parses `CARTESIAN_POINT` / `TRIMMED_CURVE`, checks the STEP SHA-256, and scales millimetres to metres. `classify_segments` labels floor ropes by the drawing $y = \pm(21.45 \pm (0.85 + 0.26k))$ lattice, long high members as portal/handrail ropes, short transverse members as `portal_or_beam`, and long inter-deck members as `cross_passage`.

Experiment. Raw counts: `floor_rope` 43 200, `portal_or_beam` 4 719, `portal_rope` 356, `cross_passage` 0, `short_other` 91 618. No single segment has $\Delta y \geq 15$ m, because each 49.655 m passage is tessellated into ≈ 1.7 m pieces. At every drawing passage X, node Y already spans $[-24.3, 24.3]$ m. Detection therefore uses Y-span, not a single long beam. Unclassified `short_other` is dropped from the coarsened deck. Longitudinal chains keep drawing stations and a 12 m target spacing.

Outlook. Portal-rope classification remains incomplete after coarsening (227 elements). That bound is reported in §9 and is not hidden by the 142 frame count.

5 Coordinate and topology gates (N07/N09)

Related work. A previous pull request shipped the pipeline and left `artifacts/` empty. An empty directory cannot be a passed coordinate gate. Topology counts must be recovered from the mesh against drawing lists, with insertions booked rather than invented silently.

Method. `infer_x_transform` scores identity, minus-$x_{\min}$, and minus raw chainage. High-Z histogram modes sit near portal clusters (≈ 700 m and ≈ 3023 m), not on the saddles. Alignment is therefore confirmed by local Z_{p90} within 12 m of $x = 666.679$ and $x = 2953.321$. Passages: 21 drawing stations $(3+13+3+2)$. Portal frames: 71 stations $\times$ 2 decks $= 142$. The Chinese unit is U+6980 (classifier for frames); U+69EC is a different character and is not used.

Table 1: Coordinate and topology gates read from `coord_gate.json`.

Check	Result	Evidence
Units	PASS	mm STEP $\rightarrow$ m nodes; $x_{\max} = 4270.609 < 20\,000$
Transform	PASS	identity, shift $= 0$
Saddle Z	PASS	north $Z_{p90} = 330.78$ m, south 324.85 m vs 340.6 m
Two decks	PASS	Y medians $+20.6$ / -20.6 m vs ± 21.45
Formed sag	PASS	geometry 214.18 m vs 227.30 m, $\lvert\Delta\rvert = 13.12$ m
No $x_{\min}$ shift in `.inp`	PASS	node $x_{\min} = 0$ equals geometry
Passages	PASS	21/21, insertions 0
Portal frames	PASS	142/142, insertions 0

Experiment. Twenty-one passages close by Y-span at every DRW-B station (Table 2). Portal frames close station-by-station (Table 5): 11 north-span, 41 main-span, 11 south 717 m, and 8 south 503 m stations, each hit on both decks. Missing stations: 0. Inserted drawing beams: 0.

Outlook. The 13.12 m sag residual is inside the 15 m gate and is not a formed-line fit. Histogram modes must not be used as saddle evidence in later runs.

Table 2: Twenty-one cross-passages (DRW-B dimension chains).

No.	Label	x (m)	Chainage
1	P01	159	K17+035.000
2	P02	330	K17+206.000
3	P03	501	K17+377.000
4	P04	838	K17+714.000
5	P05	1009	K17+885.000
6	P06	1180	K18+056.000
7	P07	1351	K18+227.000
8	P08	1504	K18+380.000
9	P09	1657	K18+533.000
10	H1	1810	K18+686.000
11	H2	1963	K18+839.000
12	H3	2116	K18+992.000
13	H4	2269	K19+145.000
14	H5	2440	K19+316.000
15	H6	2611	K19+487.000
16	H7	2782	K19+658.000
17	H8	3138.5	K20+014.500
18	H9	3318.5	K20+194.500
19	H10	3498.5	K20+374.500
20	H11	3843	K20+719.000
21	H12	4014	K20+890.000

6 Materials, disjoint anchors, and cable seed (N08–N10)

Related work. Theory v1.2 §4 forbids using a floor-rope anchor as a portal-rope anchor. The two families differ by 21 m at the north end and by 15.332 m at the south end.

Method. Rope $E = 1.20 \times 10^{11}$ Pa, floor/portal $A = 1400.42 \times 10^{-6}\,\mathrm{m}^2$, $\mu = 12.038$ kg/m (CALC-INPUT / DRW-A). Steel $E = 2.06 \times 10^{11}$ Pa, $\rho = 7850\,\mathrm{kg/m^3}$. Horizontal seed

$$H = \frac{wL^2}{8h}, \qquad h = 227.300 \text{ m}, \qquad L = 2286.642 \text{ m}.$$

Per deck, $w_{\text{floor}} = 2.766$ kN/m and $w_{\text{portal}} = 0.709$ kN/m give $H_{\text{floor}} = 7.954$ MN and $H_{\text{portal}} = 2.039$ MN. Axial seeds are $\sigma_{\text{floor}} = 3.549611 \times 10^8$ Pa and $\sigma_{\text{portal}} = 2.426295 \times 10^8$ Pa. L0 wave frequency $f_1 = \sqrt{g/32h} = 0.036719$ Hz is a magnitude gate only.

Floor and portal anchors use different node sets and two `*BOUNDARY` cards. A third card holds saddles and nominal ends.

Table 3: Disjoint anchor families. North physical anchors lie outside the STEP.

Family	Station	Target x (m)	Selected x_{mean}	Mode
Floor N	K16+852.105	−23.895	0.000	STEP north-end proxy
Floor S	K21+086.368	4210.368	4209.985	matched, $\Delta = 0.38$ m
Portal N	K16+831.091	−44.909	46.358	STEP north-end proxy
Portal S	K21+101.700	4225.700	4221.093	matched, $\Delta = 4.61$ m

Experiment. Intersection of N_FLOOR_ANCHOR (312 nodes) and N_PORTAL_ANCHOR (16 nodes) is empty. Three *BOUNDARY cards are present. South families remain 11.1 m apart in x.

Outlook. The deck does not invent nodes at negative x. A later STEP that includes the north anchorage must still keep the four families separate.

7 Load cases (N11)

Related work. CalculiX does not inherit *DLOAD/*CLOAD across steps. Each step must restate gravity and the active package.

Method. Per-deck packages are split equally onto 16 explicit floor ropes so that $\sum_i w_{\mathrm{rope}} L_i = w_{\mathrm{deck}} L_{\mathrm{deck}}$.

Table 4: Load-case resultants close against wL on the coarsened floor mesh.

Case	Deck w	Per-rope w	Resultant
Extra dead	877.16 N/m	54.82 N/m	9.019 MN
Personnel	8.400 kN/m	525 N/m	86.371 MN
Wind $+Y$	0.50 kN/m	31.25 N/m	5.141 MN

Experiment. Floor truss length in the coarsened mesh is 164 516 m (32 lines including sag and residual over-classified floor segments; $\approx$ 5141 m/line versus $\approx$ 4300 m expected). LC-FREQ requests 20 eigenvalues and does not contain 0.0296 Hz.

Outlook. The 20 % floor-line excess is a classification bound, not a live-load invention. It remains visible after the 21/142 topology audit closes.

8 Solver deck and the CalculiX 2.21 initial-stress gate (N12–N14)

Related work. CalculiX CrunchiX 2.21 (§6.11.7 and §7.76 of the User's Manual) defines *INITIAL CONDITIONS, TYPE=STRESS before the first *STEP. If USER is not specified, each data line is

element number; integration point number; S_{xx}, S_{yy}, S_{zz}, S_{xy}, S_{xz}, S_{yz}

in the global rectangular frame, as second Piola–Kirchhoff components. Abaqus-style ELSET plus a single axial stress is not that contract.

Method. The writer currently emits

```
*INITIAL CONDITIONS, TYPE=STRESS
E_FLOOR_ROPE, 3.549611e+08
E_PORTAL_ROPE, 2.426295e+08
```

at lines 90863–90865 of the frozen deck. That is ELSET + uniaxial. The hash 82548e6a is frozen: the file is not patched. CalculiX is run only on a copy under /tmp/ccx-82548e6a/. The original is re-hashed before and after.

Experiment. A lexical audit (`artifacts/ic_format_audit.json`) classifies both rows as `elset_plus_uniaxi` and `ccx_2_21_legal=false`. The formatted floor stress `3.549611e+08` equals the writer's $\sigma_{\text{floor}} = H_{\text{floor}}/(16A)$.

CalculiX 2.21 (`ccx -v`: "This is Version 2.21") was executed on the copy. The solver printed

```
*ERROR reading *INITIAL CONDITIONS. Card image:
       E_FLOOR_ROPE,3.549611E+08
...
*ERROR in calinput: at least one fatal
       error message while reading the
       input deck: CalculiX stops.
```

and exited 201. Observed artefacts: no `job.frd`; `job.dat` has 0 bytes; `job.sta` and `job.cvg` contain the printed table headers only. Wall-clock time on this host was 0.738–0.815 s, matching the given bound < 1 s. The banner "number of:" block is explicitly "estimated upper bounds" issued while reading the deck; no assembled equation count appears. The original file hash after the run is still `82548e6a...276ab6da`.

These facts were first given as an external solve on the same hash and were then reproduced locally. Both records are stored in `artifacts/ccx_run_82548e6a.json` and the copied logs `ccx_82548e6a.stdout.txt`, `.sta`, `.cvg`, `.dat`.

Outlook. A later deck may expand each tension element into an eight-field stress row without claiming that this frozen hash already did so. Expanding the card is a new artefact with a new hash. This paper does not perform that rewrite.

9 Zhangjinggao results

Related work. The scientific object is one published centre-line STEP of one construction catwalk, not a generic benchmark.

Method. Results are read back from the frozen deck, the gate JSON, the 142-frame ledger, and the CalculiX 2.21 logs. No displacement, reaction, or frequency from a completed step is reported.

Experiment.

1. Coordinate convention holds without subtracting $x_{\min}$.
2. Topology reconcile: 21 passages, 142 portal frames, both from STEP evidence, insertions 0 and 0.
3. Anchor families are lexically and set-wise separate ($312 \cap 16 = \emptyset$).
4. Independent sag seed uses 227.300 m; 255.56 m is absent.
5. All 26 automated gates PASS (`artifacts/coord_gate.json`).
6. Unit tests `test_coord_gate`, `test_write_inp`, `test_reconcile`, `test_audit_frozen_deck` PASS without rewriting the deck.
7. CalculiX 2.21 parse of `TYPE=STRESS` fails as specified; hash unchanged.

Role counts in the written deck: `floor_rope` 25 299, `portal_or_beam` 4 719, `portal_rope` 227, `cross_passage` 21, `handrail_rope` 34, `longitudinal_other` 17.

Outlook. The supported claim is the existence of a hashed, coordinate-gated, topology-reconciled deck plus a documented solver-parse failure. It is not a fourteen-mode reproduction.

10 Discussion and conclusions

Related work. Pull request #18 shipped an empty `artifacts/` directory. Pull request #19 first deposited the hashed deck. This run adds the 142-frame ledger, the CalculiX 2.21 parse record, and a complete paper, without changing `82548e6a`.

Method. Limits that remain visible in the evidence are stated as bounds, not as hidden FAIL marks. Hard gates (identity transform, 21/142, disjoint anchors, complete keywords, frozen hash, forbidden sources) all close.

Experiment. Bounds:

- North anchors are STEP-end proxies, not $K16 + 852$ / $K16 + 831$.
- Portal-rope classification is incomplete (227 coarsened elements); south portal nodes come from `portal_or_beam` at $x = 4221.093$.
- Geometric sag 214.18 m versus 227.30 m is inside the 15 m gate but is not a formed-line fit.
- Floor-line length is about 20 % above a four-span catenary estimate.
- The stress card is not CalculiX-legal. The solver never assembled a system. No spectrum exists.

Outlook. The next scientific step is a new hashed deck whose `TYPE=STRESS` rows are element number + integration point + six stresses, followed by a real nonlinear static solve, and only then a TARGET-FREQ comparison. That step is outside this frozen hash.

Reproduction

```
python3 catwalk-fem/tests/test_coord_gate.py
python3 catwalk-fem/tests/test_write_inp.py
python3 catwalk-fem/tests/test_reconcile.py
python3 catwalk-fem/tests/test_audit_frozen_deck.py
python3 catwalk-fem/pipeline/audit_frozen_deck.py
sha256sum -c catwalk-fem/artifacts/zjg_catwalk_coarsened.inp.sha256
pdflatex -interaction=nonstopmode zjg_catwalk_agentic_fea.tex
```

Do not add the 77 MB STEP to git. Do not feed `isolated/TARGET-FREQ.json` to the writer or the solver. Do not rewrite `zjg_catwalk_coarsened.inp`.

Data availability

All artefacts cited here live under `catwalk-fem/artifacts/` on branch `cursor/catwalk-main-deck-gate-f23d`. The frozen deck hash is `82548e6a3bd2612b6b39a08c313402b32a1961af6eba018158267906276ab6da`. Self-evaluation traces are in `catwalk-fem/eval/GROK_SELF_EVAL.md`.

References

[1] G. Dhondt, *CalculiX CrunchiX USER'S MANUAL version 2.21*, 2023, §6.11.7 and §7.76.

[2] Zhangjinggao catwalk fourteen-mode independent theory archive, v1.2, `catwalk-theory/thirteen-mode-models`, 2026-08-08.

[3] `catwalk-fem/SKILL.md` and `catwalk-fem/PLAN.md`, run `catwalk-main-deck-gate-f23d`.

[4] Release `catwalk-attachment23-v2.0-s10-20260716`, `cw_S10_0716t050342_a4_centerline.step`, SHA-256 `d03d01e3...763344`.

A Station-by-station 142-frame ledger

The following table is generated from `artifacts/portal_142_ledger.json` by `pipeline/audit_frozen_deck.p` Every station is present on both decks. The counting unit is frames (Chinese U+6980), not U+69EC.

Table 5: Station-by-station reconciliation of 142 portal frames (71 stations × 2 decks). Unit: frames (CJK U+6980), not the look-alike U+69EC.

No.	x (m)	Chainage	Span	Up	Dn	n_{up}	n_{dn}	OK
1	45	K16+921.000	N 660 m	Y	Y	5	5	Y
2	102	K16+978.000	N 660 m	Y	Y	8	8	Y
3	159	K17+035.000	N 660 m	Y	Y	8	8	Y
4	216	K17+092.000	N 660 m	Y	Y	31	35	Y
5	273	K17+149.000	N 660 m	Y	Y	8	8	Y
6	330	K17+206.000	N 660 m	Y	Y	8	8	Y
7	387	K17+263.000	N 660 m	Y	Y	32	35	Y
8	444	K17+320.000	N 660 m	Y	Y	9	9	Y
9	501	K17+377.000	N 660 m	Y	Y	8	8	Y
10	558	K17+434.000	N 660 m	Y	Y	34	38	Y
11	615	K17+491.000	N 660 m	Y	Y	9	9	Y
12	724	K17+600.000	Main 2300 m	Y	Y	8	8	Y
13	781	K17+657.000	Main 2300 m	Y	Y	8	8	Y
14	838	K17+714.000	Main 2300 m	Y	Y	8	8	Y
15	895	K17+771.000	Main 2300 m	Y	Y	24	29	Y
16	952	K17+828.000	Main 2300 m	Y	Y	8	8	Y
17	1009	K17+885.000	Main 2300 m	Y	Y	9	9	Y
18	1066	K17+942.000	Main 2300 m	Y	Y	25	30	Y
19	1123	K17+999.000	Main 2300 m	Y	Y	8	8	Y
20	1180	K18+056.000	Main 2300 m	Y	Y	8	8	Y
21	1237	K18+113.000	Main 2300 m	Y	Y	25	26	Y
22	1294	K18+170.000	Main 2300 m	Y	Y	8	8	Y
23	1351	K18+227.000	Main 2300 m	Y	Y	8	8	Y
24	1402	K18+278.000	Main 2300 m	Y	Y	45	50	Y
25	1453	K18+329.000	Main 2300 m	Y	Y	10	10	Y
26	1504	K18+380.000	Main 2300 m	Y	Y	10	10	Y
27	1555	K18+431.000	Main 2300 m	Y	Y	45	50	Y
28	1606	K18+482.000	Main 2300 m	Y	Y	10	10	Y
29	1657	K18+533.000	Main 2300 m	Y	Y	10	10	Y

Table 5: 142 portal frames (continued)

No.	x (m)	Chainage	Span	Up	Dn	n_{up}	n_{dn}	OK
30	1708	K18+584.000	Main 2300 m	Y	Y	44	49	Y
31	1759	K18+635.000	Main 2300 m	Y	Y	10	10	Y
32	1810	K18+686.000	Main 2300 m	Y	Y	10	10	Y
33	1861	K18+737.000	Main 2300 m	Y	Y	45	50	Y
34	1912	K18+788.000	Main 2300 m	Y	Y	10	10	Y
35	1963	K18+839.000	Main 2300 m	Y	Y	10	10	Y
36	2014	K18+890.000	Main 2300 m	Y	Y	45	50	Y
37	2065	K18+941.000	Main 2300 m	Y	Y	10	10	Y
38	2116	K18+992.000	Main 2300 m	Y	Y	10	10	Y
39	2167	K19+043.000	Main 2300 m	Y	Y	45	50	Y
40	2218	K19+094.000	Main 2300 m	Y	Y	10	10	Y
41	2269	K19+145.000	Main 2300 m	Y	Y	10	10	Y
42	2326	K19+202.000	Main 2300 m	Y	Y	29	32	Y
43	2383	K19+259.000	Main 2300 m	Y	Y	8	8	Y
44	2440	K19+316.000	Main 2300 m	Y	Y	8	8	Y
45	2497	K19+373.000	Main 2300 m	Y	Y	30	33	Y
46	2554	K19+430.000	Main 2300 m	Y	Y	9	9	Y
47	2611	K19+487.000	Main 2300 m	Y	Y	8	8	Y
48	2668	K19+544.000	Main 2300 m	Y	Y	32	37	Y
49	2725	K19+601.000	Main 2300 m	Y	Y	8	8	Y
50	2782	K19+658.000	Main 2300 m	Y	Y	9	9	Y
51	2839	K19+715.000	Main 2300 m	Y	Y	32	37	Y
52	2896	K19+772.000	Main 2300 m	Y	Y	8	8	Y
53	3018.5	K19+894.500	S 717 m	Y	Y	8	8	Y
54	3078.5	K19+954.500	S 717 m	Y	Y	9	9	Y
55	3138.5	K20+014.500	S 717 m	Y	Y	8	8	Y
56	3198.5	K20+074.500	S 717 m	Y	Y	8	8	Y
57	3258.5	K20+134.500	S 717 m	Y	Y	8	8	Y
58	3318.5	K20+194.500	S 717 m	Y	Y	8	8	Y
59	3378.5	K20+254.500	S 717 m	Y	Y	9	9	Y
60	3438.5	K20+314.500	S 717 m	Y	Y	8	8	Y
61	3498.5	K20+374.500	S 717 m	Y	Y	9	9	Y
62	3558.5	K20+434.500	S 717 m	Y	Y	8	8	Y
63	3618.5	K20+494.500	S 717 m	Y	Y	8	8	Y
64	3729	K20+605.000	S 503 m	Y	Y	8	8	Y
65	3786	K20+662.000	S 503 m	Y	Y	8	8	Y
66	3843	K20+719.000	S 503 m	Y	Y	8	8	Y
67	3900	K20+776.000	S 503 m	Y	Y	23	24	Y
68	3957	K20+833.000	S 503 m	Y	Y	8	8	Y
69	4014	K20+890.000	S 503 m	Y	Y	8	8	Y
70	4071	K20+947.000	S 503 m	Y	Y	22	24	Y
71	4128	K21+004.000	S 503 m	Y	Y	8	8	Y

B CalculiX 2.21 transcript (abridged)

The complete stdout is `artifacts/ccx_82548e6a.stdout.txt`. The fatal lines are:

```
*ERROR reading *INITIAL CONDITIONS. Card image:
```

```
       E_FLOOR_ROPE,3.549611E+08
*ERROR in calinput: at least one fatal
       error message while reading the
       input deck: CalculiX stops.
```

`job.sta`:

```
SUMMARY OF JOB INFORMATION
  STEP       INC      ATT  ITRS     TOT TIME     STEP TIME      INC TIME
```

`job.cvg` likewise contains only the convergence-table header. `job.dat` is empty. `job.frd` does not exist.